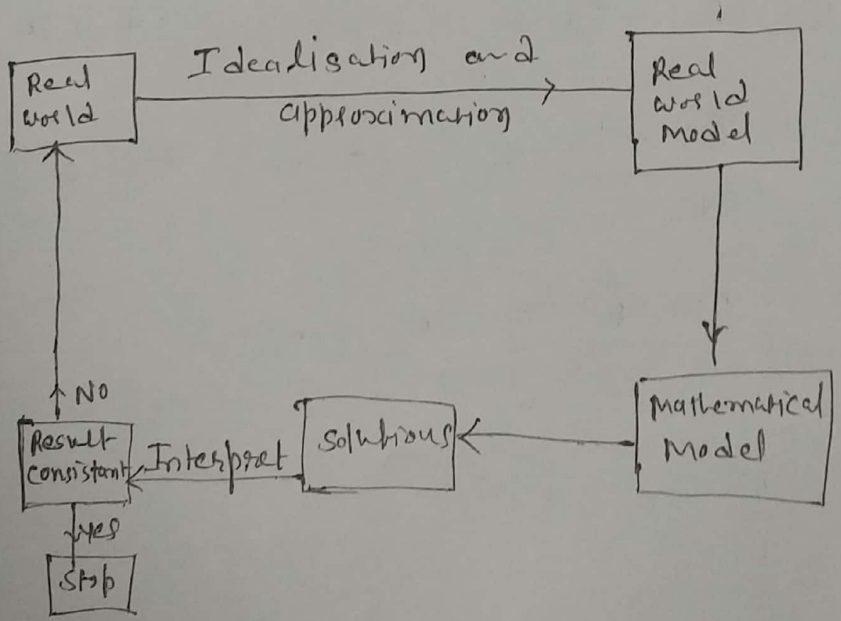
Lecture Note: Unit-II

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* Mathematical Model:

Mathematical Modelling essentially consist of translating real world problems into mathematical problems, solving the mathematical problems and interpreting these solutions in the lesser language of real world



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Model-I Newton's Law of Cooling

According to Newton's Law of cooling, the rate of change of temperature of a body is proportional to the difference between the temperature of the body and temperature of the surrounding medium.

Let T be the temperature of the body.

and T_s be the temperature of the surrounding medium

Let $T(t)$ be the temperature at time t

and $T(t+\Delta t)$ be the temperature at time $t+\Delta t$

Hence,

$$(T(t+\Delta t) - T(t)) \propto (T - T_s).\Delta t + O(\Delta t)$$

$$\Rightarrow \frac{(T(t+\Delta t) - T(t))}{\Delta t} \propto (T - T_s) + O(\Delta t)$$

$$\Rightarrow \lim_{\Delta t \to 0} \left[\frac{(T(t+\Delta t) - T(t))}{\Delta t} \right] \propto (T - T_s)$$

$$\Rightarrow \frac{dT}{dt} \propto (T - T_s)$$

$$\Rightarrow \frac{dT}{dt} = K(T - T_s) \qquad K < 0$$

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Solutions

$$\frac{dT}{T-T_s} = k\,dt$$

$$\Rightarrow \int \frac{dT}{T-T_s} = \int k\,dt + G$$

$$\Rightarrow \log(T-T_s) = kt + G$$

$$\Rightarrow T - T_s = e^{kt+G}$$

$$\Rightarrow T - T_s = A\,e^{kt} \quad [\because A = e^{G}] \quad \text{— (1)}$$

let, at $t=0$ $T(0) = T_0$ be the initial condition

$$\Rightarrow T_0 - T_s = A\,e^{0}$$

$$\Rightarrow T_0 - T_s = A \quad \text{— (2)}$$

from (1) and (2)

$$T - T_s = (T_0 - T_s)\,e^{kt}$$

$$\Rightarrow \boxed{T = T_s + (T_0 - T_s)\,e^{kt}} \qquad k<0$$

This shows that the excess of the temperature of the body over that of the surrounding medium decays exponentially

Example-1 According to Newton's law of cooling, the rate at which a substance cools in moving air is proportional to the difference between the temperature of the substance and that of air. If the temperature of the air is 40°C and the substance cools from 80°C to 60°C in [illegible] minute.

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in 20 minutes. What will be the temperature of the body after 40 minutes from the original?

Sol $T_s = 40\,^\circ C$

Hence, $\frac{dT}{dt} = k(T - 40)$

$T = T_s + (T_0 - T_s)e^{kt} \Rightarrow T = T_s + (40 - T_s)e^{kt}$

$T = 40 + (T_0 - 40)e^{kt}$

at $t = 0$ $T = 80$

$\Rightarrow 80 = 40 + (T_0 - 40)e^0$

$\Rightarrow \boxed{80 = T_0}$

when $t = 20$ $T = 60$

$\Rightarrow 60 = 40 + (80 - 40)e^{20k}$

$\Rightarrow 20 = 40e^{20k}$

$\Rightarrow \frac{1}{2} = e^{20k}$

$\Rightarrow 20k = \log(1/2)$

$\Rightarrow k = \frac{1}{20}[\log 1 - \log 2]$

$\Rightarrow k = -\frac{1}{20}\log 2$

at $t = 40$

$T = 40 + (80 - 40)e^{(-\frac{1}{20}\log 2)\times 40}$

$= 40 + (40)\left[e^{-\frac{1}{20}\log 2 \times 40}\right]e^{-2\log 2}$

$= 40 + 40\left[\frac{1}{2^{1/20 \times 40}}\right]e^{\log 2^{-2}}$

$T = 40 + 40 \times 2^{-2}$

$= 40 + \frac{40}{4}$

$= 40 + 10$

$\boxed{T = 50}$

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Example: Water at temperature 100°C cools in 10 minutes to 88°C in a room of temperature 25°. Find the temperature of water after 20 minutes.

$T_0 = 100°C$

$T_s = 25°C$

$t = 10, \quad T = 88°C$

$$\Rightarrow T = T_s + (T_0 - T_s)e^{kt}$$

$$\Rightarrow 88 = 25 + (100 - 25)e^{k10}$$

$$\Rightarrow 88 - 25 = 75\, e^{k10}$$

$$\Rightarrow \frac{63}{75} = e^{10k}$$

$t = 20 \qquad T = ?$

$$\Rightarrow T = T_s + (T_0 - T_s)e^{kt}$$

$$\Rightarrow T = 25 + (100 - 25)e^{20k}$$

$$= 25 + 75\left(e^{10k}\right)^2$$

$$T = 25 + 75\left(\frac{63}{75}\right)^2$$

$$\boxed{T = 77.92°C}$$

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Example If the temperature of the air is 30 °C and the substance cools from 100 °C to 70 °C in 15 minute, find when the temperature will be 40 °C [52.5 minute]

Example - A body where temperature T is initially 300 °C is placed in a large block of ice. Find its temperature at the end of 2 and 3 minutes?

Model - II Simple Compartment Model

Let a vessel contain a volume V of a solution with concentration C(t) of a substance at time t. Let a solution with constant concentration C_0 in an overhead tank enter the vessel at a constant rate R and after mixing ~~thoroughly~~ thoroughly with the solution in the vessel, let the mixture with concentration C(t) leave the vessel at the same rate R, so that the volume of the solution in the vessel remains V.

Using the principle of continuity

Let C(t) be the concentration at t and C(t+Δt) be the concentration at t+Δt

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we get,

$$V\,C(t+\Delta t) - V C(t) = RG\Delta t - R C(t)\Delta t + O(\Delta t)$$

$$\Rightarrow V\left[\frac{C(t+\Delta t) - C(t)}{\Delta t}\right] = RG - RC(t) + O(\Delta t)$$

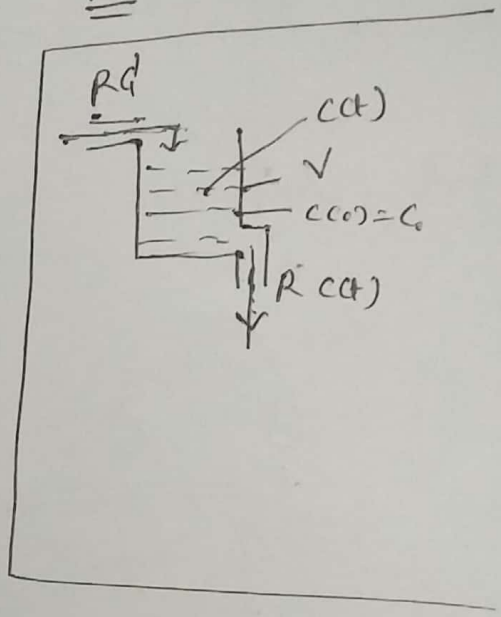
$$\Rightarrow V\left[\lim_{\Delta t \to 0} \frac{C(t+\Delta t) - C(t)}{\Delta t}\right] = RG - RC(t)$$

$$\Rightarrow V\frac{dC}{dt} = RG - RC(t)$$

$$\Rightarrow \frac{dC}{dt} = \frac{R}{V}G - \frac{R}{V}C(t)$$

$$\Rightarrow \boxed{\frac{dC}{dt} + \frac{R}{V}C(t) = \frac{R}{V}G}$$

Fig



~~at $t=0$, $C(0) = C_0$~~

Solution

$$I.F = e^{\int \frac{R}{V} dt} = e^{\frac{R}{V}t}$$

Hence, $$C(t)\, e^{\frac{R}{V}t} = \int \frac{R}{V} G\, e^{\frac{R}{V}t} dt + A$$

$$\Rightarrow C(t)\, e^{\frac{R}{V}t} = \frac{R}{V} G \left[\frac{e^{\frac{R}{V}t}}{\frac{R}{V}}\right] + A$$

$$\Rightarrow C(t)\, e^{\frac{R}{V}t} = G\, e^{\frac{R}{V}t} + A$$

$$\Rightarrow C(t) = G + A\, e^{-\frac{R}{V}t} \quad \text{——— (1)}$$

at $t=0$, $C(0) = C_0$

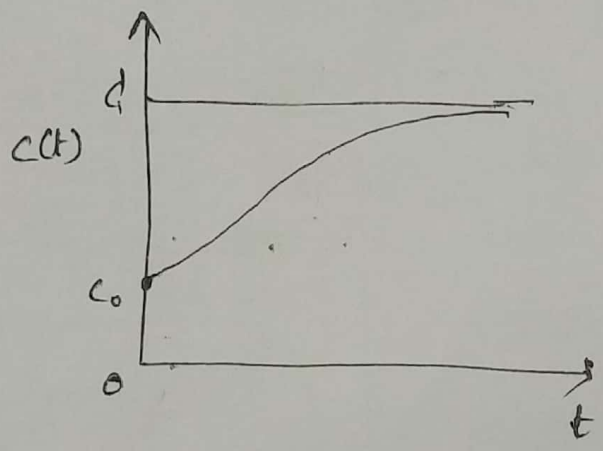
$$\therefore \boxed{A = C_0 - G} \quad \text{——— (2)}$$

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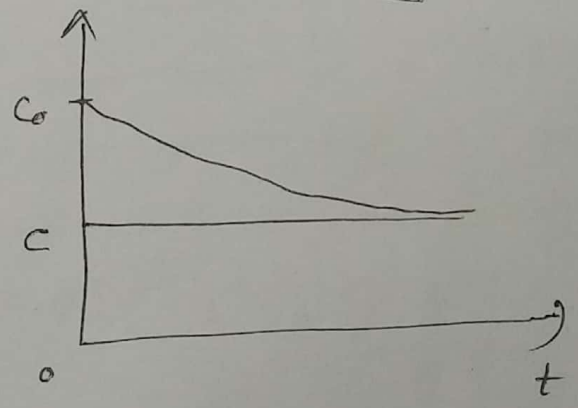
from ① and ② we get,

$$C(t) = G + [C_0 - G]\, e^{-\frac{R}{V}t}$$

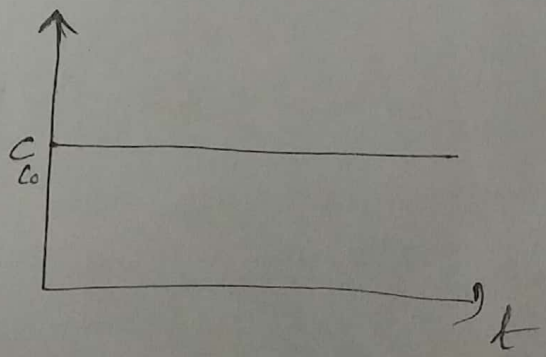
when ① $G > C_0$ [Graphical Analysis]



② $G < C_0$



③ $C = C_0$



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Example:-1

A 2000 \ liters tank of waters initially contains 20 kg of dessolved salt. A pipe brings salt-solution with concentration 0.04 kg/liters in to the tank at the rate 2 liters/second and a second pipe exies away the excess solution at the same rate. Calculate concentration of salt at any time with assumption that tank is well mixed.

Sol^n

$$V = 2000 \text{ liters}$$

$$C_0 = \frac{20}{2000} = 0.01 \text{ kg/liters}$$

$$R = 2 \qquad G = 0.04 \text{ kg/liters}$$

$$C(t) = Gt + [C_0 - G]\, e^{-R/V\, t}$$

$$\Rightarrow C(t) = 0.04 + [0.01 - 0.04]\, e^{\frac{2}{2000} t}$$

$$\boxed{C(t) = 0.04 - 0.03\, e^{0.001t}}$$

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Example:

A 8000 liters tank of water initially contains 5 kg of dissolved potassium. A pipe brings a potassium solution with concentration 0.03 kg/liter in to the tank at the rate of 4 l/s and a second pipe dries away the excess solution with the same rate. Find the concentration of potassium at any time t in the tank

→ $V = 8000$

$C_0 = 5/8000 = 0.000625$ kg/liter

$C' = 0.03$ kg/liter

$R = 4$

$$C(t) = C' + [C_0 - C'] e^{-R/V t}$$

$$\Rightarrow \boxed{C(t) = 0.03 + [0.000625 - 0.03] e^{-4/8000 t}}$$

~~Example~~ :

Example: A 4000 liter tank of water ~~was~~ initially contains 4 kg of dissolved salt. A pipe brings a salt solution with concentration 0.02 kg/liter in to the tank at the rate of 2 l/s and a second pipe dries away the excess solution with the same rate. then calculate concentration